Inorganic CUME on Materials Chemistry
September 9th-2016
Answer any 5 questions (but only 5)

1. In the standard notation of closest-packing of spheres, the letters A, B and C refer to close-packed **single** layers (See Figure 1 below). These layers can be stacked on top of each to form 3 dimensional structures. Which of the following sequences describe **closest-packing in 3 dimensions**, and which do not. Explain why for each.

   (i) (5points)ABCABC....
   (ii)(5points)ABBABABBA....
   (iii)(10points)ABACABACABAC....

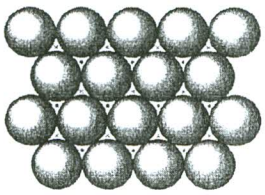

Figure 1. Top view of single layer (A, B or C) of close-packed spheres.

2. (20 points) Briefly explain the working principle of X-ray diffraction (XRD). What is the Bragg's Law? How to use Bragg's Law to calculate the lattice constants in a cubic structure? Give an example of using the XRD for materials research.

3. (20 points) A copper foil substrate (about 20 μm thick) was coated with a very thin layer (less than 200 nm) of poly (2-vinyl pyridine) (P2VP) (see the molecular structure below in Figure 2) whose thickness cannot be easily measured by means of the scanning electron microscope (SEM). Please list at least two alternative approaches (10 points for each method) that can be used to probe the thickness of P2VP on the copper foil.

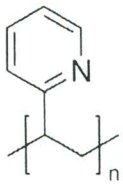

Figure 2. Molecular structure of P2VP

4. (20 points) Calculate the number of atoms in each of the four different unit cells below (Figure 3).

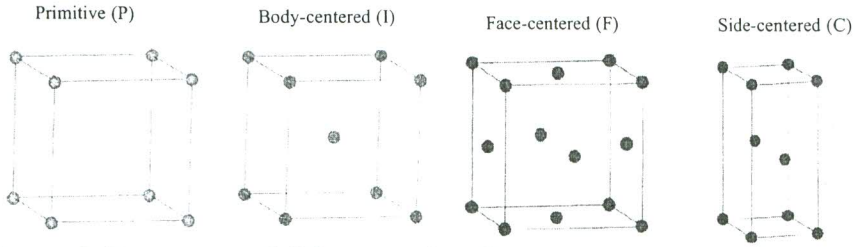

Figure 3. Four types of different unit cells

5. (20 points) Why diamond is electrically insulating and hard, while graphite is electrically conductive and slippery, although they are both made of pure carbon? Which one, diamond or graphite, is the most stable form at standard thermodynamic conditions? What is the difference between quartz and glass in structural chemistry?

6. Copper hexacyanoferrate, $KCuFe(CN)_6$, has the same open framework structure of Prussian Blue hexacyanometalate, nominally, $KFe_2(CN)_6$. Both have been recently found workable as battery electrodes in inexpensive aqueous electrolytes, which are promising for large-scale energy storage systems (Figure 4). Answer the following questions:

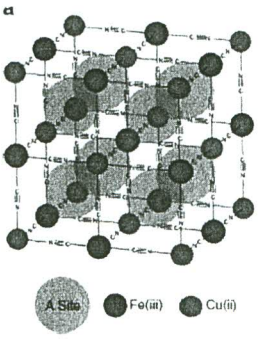

Figure 4. Structure of $KCuFe(CN)_6$. $KCuFe(CN)_6$ has the Prussian Blue crystal structure, in which Cu and Fe are linked by CN ligands, forming a face-centered cubic structure. Each of the eight subcells of the unit cell contains a zeolitic 'A site ' that can be occupied by H_2O or hydrated alkali cations e.g. K^+.

(Source: Nature Communications 2, 550 (2011)

(a) (10 pints) What are the coordination numbers of Cu and Fe in $KCuFe(CN)_6$?
(b) (10 points) On the insertion of extra K+ ions into the Prussian blue, there is a change in color from deep blue to colorless. Why is that?

7. (20points) One mole of $KCuFe(CN)_6$ electrochemically reacts with up to one mole of K^+ ions. The equation is written as: $KCuFe(CN)_6 + K^+ + e^- = K_2CuFe(CN)_6$. Calculate the theoretical capacity of anhydrous $KCuFe(CN)_6$. [K: 39 g/mol; Cu:64 g/mol; Fe:56 g/mol; C:12 g/mol; N:14 g/mol.

8. (20 points) Give a formula (or sketch) for each common organic ligand that can act as a) one-electron donor, b) two-electron donor, c) three-electron donor, and d) four-electron donor.

9. (20 points) Iron pentacarbonyl, like many metal carbonyl compounds, is photochemically very active. Traditionally, it is irradiated in glacial acetic acid to form a product in nearly quantitative yield. Deduce the product and indicate how you would use IR spectroscopy to verify the product.

10. (20 points) $CpMn(CO)_3$ is among the most thermally stable organometallic compounds known to date. It does not react even if being refluxed in triphenylphosphine for several hours. Propose two reactions that take advantage of the thermal stability but lead to potentially useful products. The reactions should be generally applicable to other similar synthetic routes. Provide some text in support of your choice.